

Experiment 01 — ExpressJS Routing, HTTP Methods and Middleware

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Start an Express server on port 3001.
- Use custom middleware to log requests.
- Read route parameters such as /students/1.
- Read query parameters such as /search?name=Rama.
- Build a URL dynamically.
- Create, read and delete student records in memory.
- app.js — complete ExpressJS example with comments and clear sections.

2. Workflow

Experiment 01 — ExpressJS Routing, HTTP Methods and Middlewar



Workflow: input → processing → data/state → output

3.1. Code pathway — app.js

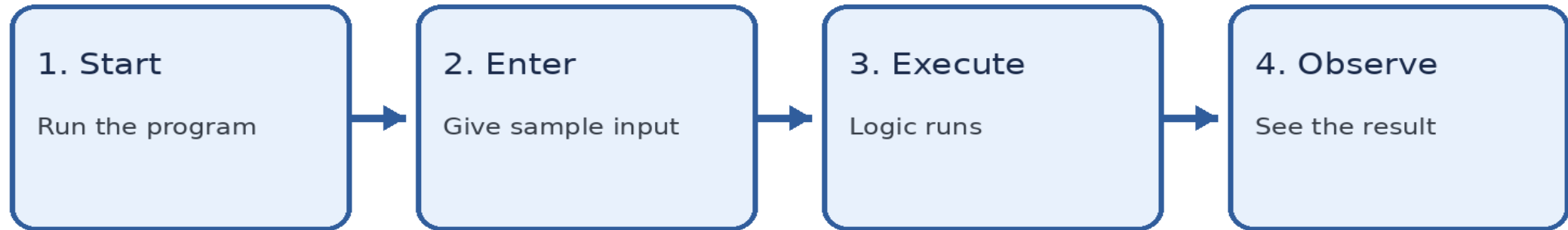
```
1 /**
2  * Experiment 01: ExpressJS Routing, HTTP Methods and Middleware
3  *
4  * What it demonstrates:
5  * - Express server setup
6  * - Custom middleware
7  * - Route parameters and query parameters
8  * - URL building
9  * - GET, POST and DELETE methods
10 */
```

3.2. Code pathway — app.js

```
11
12 const express = require('express');
13
14 const app = express();
15 const PORT = 3001;
16
17 // -----
18 // Middleware
19 // -----
20
```

4. Runtime walkthrough

Runtime walkthrough — Experiment 01 — ExpressJS Routing, HTTP Methods



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?